

Research paper

The science planning process on the Rosetta mission



Claire Vallat^{a,*,1}, Nicolas Altobelli^{b,1}, Bernhard Geiger^c, Bjoern Grieger^c, Michael Kueppers^b, Claudio Muñoz Crego^d, Richard Moissl^b, Matthew G.G.T. Taylor^e, Claudia Alexander^{f,2}, Bonnie Buratti^f, Mathieu Choukroun^f, and the RSGS Liaison scientists group

^a Telespazio VEGA PLC, European Space Astronomy Centre/ ESA, Camino Bajo del Castillo s/n, 28691 Villanueva de la Cañada, Madrid, Spain

^b European Space Astronomy Centre/ ESA, Madrid, Spain

^c Aurora Technology B.V., European Space Astronomy Centre/ ESA, Madrid, Spain

^d GMV, European Space Astronomy Centre/ ESA, Madrid, Spain

^e European Space Research and Technology Centre / ESA, Noordwijk, The Netherlands

^f Jet Propulsion Laboratory, Pasadena, United States

ARTICLE INFO

Keywords:

Rosetta

Comet

Science Planning

Operations

ABSTRACT

The Rosetta mission arrived at comet 67 P/Churyumov-Gerasimenko in Summer 2014, after more than 10 years in space. All previous mission encounters with a comet have provided a snapshot of the cometary activity at a given heliocentric distance. In contrast, Rosetta has escorted the comet nucleus for an extended period (> 2 years) at a large range of cometo-centric and heliocentric distances, which has provided exceptional and unprecedented observing conditions to study, analyse and monitor 67 P during its passage to, through and away from perihelion.

One of the biggest challenges of this mission is the development of an observation plan that adequately addresses the mission's science objectives while coping with a largely unknown and continuously evolving environment that constantly modifies the planning constraints.

The Rosetta Science Ground Segment (RSGS), in support of the Project Scientist and the Science Working Team, is in charge of translating the high level mission science objectives into a low level pointing and operations plan.

We present here the high-level science planning process adopted during the comet escort phase. We describe the main science objectives addressed along the mission lifetime, the different groups involved in the science planning, and the approach followed to translate those requirements into a viable and scientifically valid operations plan. Finally, we describe how the science planning scheme has evolved since arrival at the comet to react to the unexpected environment, largely reducing the planning lead times.

1. Rosetta mission

The European Space Agency's (ESA) Rosetta mission is carrying out a true exploration of an unknown world: It is the first rendezvous with and landing on a comet. This has led to a break-through in cometary science and unprecedented public attention for a space mission to a minor body, but it also required new challenges to be overcome for science and operations planning. The properties (including its shape) of the nucleus were not well known before Rosetta arrived, and the gas and dust activity of comet 67 P/Churyumov-Gerasimenko could only be investigated when the spacecraft was already within the coma. Throughout the mission, the spacecraft has

to deal with a large range of environmental conditions.

Specific challenges relevant to the area of science planning are:

- Environmental constraints: Rosetta is the first spacecraft to navigate within close range (tens to hundreds of kilometers) of a comet for an extended period. In addition to the gravity of the comet (the deviation from point source gravity due to the non-spherical shape is significant at close distance) and the one of the Sun, the drag force from the gas streaming away from the nucleus and solar radiation pressure also has to be considered. In particular gas drag limits the predictability of short-term trajectories and of the trajectory types that can be flown on the long-term. In addition, the dust particles

* Corresponding author.

E-mail address: cvallat@sciops.esa.int (C. Vallat).

¹ Both authors contributed equally to this paper.

² Passed away.

- emitted by the nucleus can pose a hazard for the spacecraft and instruments and further limit the trajectory options.
- Spacecraft constraints: Rosetta operates between 0.9 AU and 4.5 AU from the sun. At large heliocentric distance operations are constrained by limited power availability, while at lower distances thermal constraints limit the possible spacecraft attitudes. Attitude limitations in turn feed back into trajectory constraints through the need of regularly pointing the navigation cameras to the nucleus.
 - Lander support: One of the great achievements of the Rosetta mission is the first landing on a cometary nucleus with the Rosetta lander Philae. At the same time, there was little time available between orbit insertion and landing, and consequently the early phase of the mission focussed on preparation for and implementation of the landing. This meant that the mission related operations were prioritised over science. However, in most cases the requirements for landing site selection and related operations also supported significant scientific observations.
 - The scientific objectives of the mission change while the target comet is explored based on science and operational changes. A major example of that is the impact of changes driven by the dust environment on the operation of the spacecraft in the vicinity of the nucleus.
 - The diverse groups of international participants and instruments, some of which have divergent pointing and distance requirements, presented the need to find compromises in mission planning and data collection strategies.

Due to such constraint evolution and the experience gained, the science planning process also evolved during the mission. Before lander delivery, the trajectory and high priority observations of Rosetta orbiter instruments were pre-defined based on the needs of Philae (May – December 2014). Purely scientific observations were then added as far as resources allowed. For the phase after landing and at moderate comet activity, trajectories were planned several months in advance based on scientific requirements and predictions of activity of the comet, and observations were distributed according to scientific disciplines (December 2014 – March 2015). From April 2015 on, the operations scheme has been modified to deal with unpredictability of the comet activity and to increase flexibility in adapting to new conditions: trajectories are selected only days in advance, and chosen based on scientific high level criteria. As much as possible operations are adapted to the trajectories actually flown.

The present paper focuses on the science planning activities in the scientific phase (post lander delivery) of the mission. It is organized as follows: In Section 2 the science goals of Rosetta and their distribution over the mission are described. Section 3 describes the overall science operations planning concept and the groups involved in the science planning steps. Section 4 details the science planning process. Section 5 explains the evolution to the current planning scheme. We conclude in Section 6 with a discussion of the benefits and limitations of both schemes (before and after March 2015).

2. Master science plan

The Rosetta mission science planning follows a scientific roadmap or Master Science Plan (MSP), created by the Science Working Team (SWT, see below), and based on the high level goals outlined in the Rosetta Science Management Plan [1]. The MSP provides guidance for the prioritisation of investigation types or instrument operations by mission phase and also tracks the completion of the mission's top level science objectives.

The overarching science goal of Rosetta is to understand the physical and chemical processes that drive the cometary activity. In particular, to disentangle which constituents and characteristics originate from the formation of the comet and which ones have developed throughout the comet's history in the solar system. Such an understanding can be gained only by monitoring a certain number of physical

properties of the comet as function of the heliocentric distance over a full activity cycle (from 'dormant' inactive state to the maximum of activity and back to dormant).

The MSP is divided into three main steps: 'First Time', 'Observing the Development of Cometary Activity' and 'Comparison with Pre-Perihelion Conditions'. Additionally, the MSP also includes an 'Extension' phase. We summarize in the next paragraphs the main characteristics and scientific content of each phase.

2.1. First time

During this phase, Rosetta was 'getting acquainted' with the comet and fundamental measurements were performed for the first time. The 'First Time' phase has been conducted during December 2014 until March 2015, at a distance range of 2.95 to 2.1 Astronomical Units (AU) and while the Northern hemisphere of the nucleus was illuminated.

It is important to note that fundamental properties of the nucleus had been derived during the pre-landing phase, as part of the operational support for the Philae landing [2]. Before the 'First Times' science phase, initial values of the comet rotation period, mass, volume, density and interior structure had been derived, as well as the shape of the illuminated regions characterized to a < 50 cm scale resolution.

The overarching aspect of the 'First Times' science objectives was the characterisation of the relatively inactive nucleus and its environment as a basis for comparison to the later changes, after large scale activity onset. We list below the science objectives addressed during this 'First Time' phase (although some continue into other phases). Note that the First Times phase of the orbiter science plan included observation campaigns in direct collaboration, or complemented by, the First Science Sequence of the Philae lander.

- What is the distribution of minerals, organics and ice on the comet's surface? ['Ground-truth' provided by the measurement of Philae for investigated surface elements]
- What are the di-electrical surface properties of the nucleus? ['Ground-truth' provided by the measurement of Philae for investigated surface elements]
- What are the thermal properties/energy balance of the surface material? ['Ground-truth' provided by the measurement of Philae for investigated surface elements]
- What is the composition of the dust and gas around the nucleus at very low activity level (including isotopes)?
- Are there any icy grains outside the snow line?
- Characterise the dynamics of dust, and the dust physical properties around the nucleus at very low activity level
- Search for icy patches on the nucleus and search for any sign of activity onset
- Quantify the relative contribution of H₂O, CO₂ and CO to the low activity regime
- Observe the interaction of the solar wind with the nucleus surface (sputtering)

2.2. Observing the development of cometary activity

This second main phase of the Rosetta MSP covers a distance range from around 2.1 AU down to the perihelion at 1.25 AU, corresponding to the time between April-August 2015. During this phase, the comet activity increased and the nucleus began to become shielded from the solar wind environment as the coma fully developed. This period also coincides with the Southern equinox (May 2015) where the Southern hemisphere of the nucleus started to get illuminated after a long winter.

The main science objectives of this phase are listed below.

- Evolution of nucleus geomorphology and surface composition and thermal properties with developing activity

- Coma characterisation: distribution of H₂O, CO, CO₂ and how does this relates with surface activity.
- What are the composition, dynamics, morphology and optical properties of dust in a developed coma? How did the dust properties change compared to the populations observed at the lowest activity levels?
- How are dust composition and gas composition related? Are there dust sublimation processes?
- How are dust particles lifted from the surface and accelerated?
- How does the dust size distribution evolve in a developed coma? What is the importance of fragmentation processes?
- What is the dust-to-gas mass ratio?
- How does dust interact with the plasma formed by the ionized coma species?
- Looking for jets, how do they relate to surface features? Is the composition of jets different from the coma average?
- How does activity evolve over a selected patch of nucleus surface over a daily cycle?
- What are the differences (morphology, composition) between an active and an inactive nucleus area?
- What are the chemical reactions at play in the coma between neutral gas, ions and dust?
- How do the inner coma and solar wind interact? Can a magnetic cavity form?
- When is the peak of activity reached?

2.3. Comparison with pre-perihelion conditions

This phase covers the time from perihelion (13th August 2015) up to December 2015.

After the perihelion passage, cometary activity continued to intensify for a few weeks. The maximum of activity was predicted around the 30th of August, two weeks after perihelion, based on previous ground based observations [3] and observations during this period indicate similar behaviour for this apparition. The decline of the activity marked the beginning of the last phase of the nominal mission, with a focus based on comparing the post-perihelion measurements with pre-perihelion conditions. The comet behaviour is not expected to be symmetrical with respect to the point of highest activity ([3–5]), although initial observations of this period by Rosetta indicate some levels of symmetry. As the Rosetta mission aims at characterising the cometary activity in detail over nearly a full cycle of activity, this last phase is far from being a mere repetition of pre-perihelion measurements, and instead, is crucial at providing the full picture of the mechanisms driving cometary activity and how those affect the comet's characteristics. Last but not least, the time around perihelion passage also marked a change of season on the nucleus with a southern Spring

and Summer. Owing to the comet rotation axis tilt and the orbit's eccentricity and the close distance to the Sun, those seasons are intense and short. As a result, newly illuminated regions of the Southern hemisphere could be observed for the first time and frequent activity outburst were witnessed as regions which had been in the dark for years came into intense sunlight.

The main objectives of this phase are listed below:

- Evolution of the nucleus dynamical properties (rotation period, potential excitation states)
- Characterisation of the nucleus regions exiting the 'polar night'
- Re-surfacing (geo-morphology and composition) under the action of activity
- Temporal evolution of ion and gas composition and comparison since onset of activity
- Measure the dust-to-gas mass ratio evolution and compare with pre-perihelion conditions
- Compare the dust size distribution, dynamics, composition, morphology and compare with pre-perihelion conditions
- Coma-IMF interactions: characterizing the return to solar wind dominated conditions
- How does jet activity decline, what regions remain active, are they different from the regions observed to be active pre-perihelion?
- Effect of season on activity

2.4. Extension

This phase covers December 2015 to September 2016 and focuses on post-perihelion activity as the comet moves to larger and larger heliocentric distances. The resulting reduction in cometary activity thus provides the fullest picture of activity possible and how it changes with time. The phase facilitates enhanced context from ground-based observations, a key aspect for relating Rosetta observations to all other cometary observations, due to optimized viewing conditions during this period.

This period is also dedicated to carry out investigations that were not possible during the nominal mission, such as the inclusion of a nightside excursion. It also ensures that any goals not captured during the nominal phase of the mission are given the best chance of being addressed, in particular by taking larger risks and getting extremely close to the nucleus.

3. Overall planning process structure

The science operations planning process maps the Master Science Plan into an instrument operation timeline that fits within the instruments and spacecraft resources and constraints. The planning

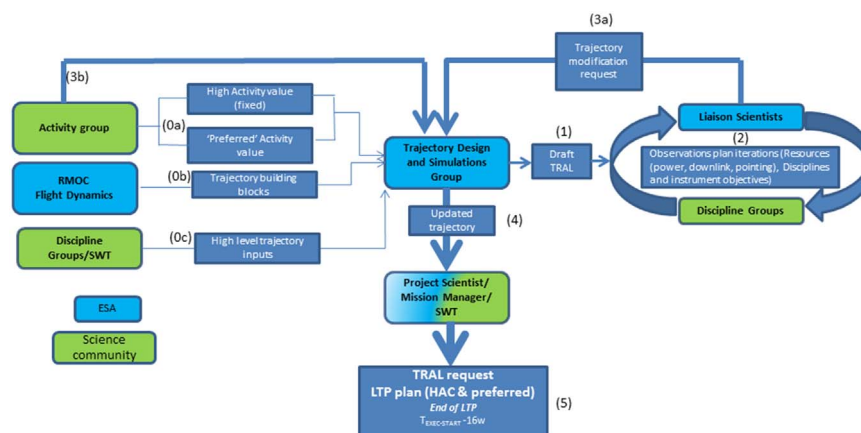


Fig. 1. Overview of the Science Planning Process until up to the end of the Long Term Planning cycle. Numbers in parenthesis show the steps order and are described in details in Section 4. Note that the trajectory segmentation activity is not shown in this scheme.

Table 1

Schematic view of the science planning process breakdown; a typical LTP period covers up to 4 months of operations (3 in this example), an MTP covers 4 weeks and an STP covers 1 week.

LTP004											
MTP011				MTP012				MTP013			
STP35	STP36	STP37	STP38	STP39	STP40	STP41	STP42	STP43	STP44	STP45	STP46

process is divided in two parts: the science planning process (see Fig. 1) followed by the actual implementation of the observations plan at pointing and commanding level. This latter step is detailed in [6].

3.1. Cycles and lead-in times

The science operations planning process during the various phases of the Rosetta mission follows three cycles: the long term (further referred to as LTP), the medium term (or assembly, MTP) and the short term planning (STP). This paper describes in details the LTP process only (Section 4), but an overview of all planning phases is given in this section. Thus there are 4 STP periods in one MTP, and up to 4 MTP periods in one LTP (Table 1). In addition, each STP is divided into two Very Short Term Planning (VSTP) cycles, one of three days and one of four days. All lead in times are worked backward from the processing times imposed by the Rosetta Mission Operations Centre (RMOC) before uplink to the spacecraft..

3.2. The long term planning

Although in practice the science planning iterations start well before that (and finishes at the end of this phase), the LTP planning cycle officially begins at start of execution minus 28 weeks. The main actors of activities related to Science Planning are presented in the following sub-sections.

3.2.1. Groups involved in the science planning

3.2.1.1. The instrument teams. All instruments on board Rosetta have been designed and developed by a so-called Principal Investigator (PI) team located in different institutes, in Europe and in the United States. Those teams are in charge of defining the scientific observations and preparing the sequences of operations of their instrument and the Principal Investigator of each team is a member of the Science Working Team.

3.2.1.2. The science working team. The Science Working Team (SWT) consists of 11 instrument Principal Investigators (PIs), 4 interdisciplinary scientists and 2 lander lead scientists representing the Philae lander PIs. The chairperson of the SWT is the ESA Project Scientist with support from his NASA counterpart. The SWT is responsible for the scientific implementation of the mission. The SWT mandates a number of tasks to other groups such as the trajectory and Activity Group or the Discipline Groups (see below) to support the science planning. It is supported on ESA side by the Rosetta Science Ground Segment (RSGS) and the Rosetta Mission Operations Centre (RMOC). For science planning, the SWT is mainly supported by the RSGS Liaison Scientists.

3.2.1.3. The (SWT) trajectory and activity group. One of the major activities of the science planning process has been the trajectory design. The SWT provides the RSGS with its requirements for the trajectory to be flown during a particular portion of the Rosetta mission. These requirements are based on the Rosetta Master Science Plan and define

the first-order interest in certain trajectory parameters, like distance to comet or phase angle at closest approach. The result is a draft trajectory. The draft trajectory is a crucial aspect of the high level science planning as it enables the assessment of the feasibility of implementing particular prime observation objectives. This is further discussed below.

The cometary activity has a significant impact on the capability of the spacecraft to fly a particular planned trajectory. To ensure that the planned trajectory could be flown (and associated observations could be carried out), an activity group was mandated by the SWT to combine cometary simulations, models, Rosetta observations and current and previous ground based observations to provide predictions of activity. These activity predictions were used for checking the validity of long term draft trajectories.

3.2.1.4. The (SWT) discipline groups. The Discipline Groups (DGs) were created as a non-ESA component of science planning in order to develop the synergies and exploit the interdependence of instruments, potentially conflicting at operational level while they pursue the same, or complementary scientific goals.

For example, dust properties can be best constrained by a combination of ‘remote sensing’ instrument campaigns (i.e. spectral measurements), sensitive to integrated quantities along their line of sight, while the local truth will be acquired by dedicated ‘in-situ’ instruments. However, the remote sensing and in-situ instruments would conflict operationally by the pointing requested: the in-situ instruments rely on pointing radially towards the comet in order to be sensitive to gas and dust emanating from the nucleus on straight trajectories, while the remote sensing instruments are most sensitive to light scattered at specific phase angles. They would also compete for the available data volume.

It was therefore important, when the DGs were created, that their setup allows for an efficient resolution of operational conflicts between instruments. This is only possible if their members are representing instrument teams sharing the same science goal, such as to foster compromises. The concept of Discipline Groups and its implementation for Rosetta follows the successful example as given by the Cassini mission [7]: four discipline groups are defined for science planning purposes around four main science themes, all defined in the Science Management Plan [1]. DG1 is dedicated to the global characterisation of the nucleus, determination of its dynamic properties, surface morphology and composition. DG2 addresses the chemical, mineralogical and isotopic composition of volatiles and refractories in the cometary nucleus. DG3 focuses on the physical properties and interrelation of volatiles and refractories in a cometary nucleus. The task of DG4 is to study the development of cometary activity and the processes in the surface layer of the nucleus and in the inner coma (dust-gas interaction). Each DG is chaired by a RSGS Liaison Scientist (see 3.2.1.6) and has as members at least one representative of each instrument team who has been empowered by their PI to participate to negotiations on operations for their instrument.

Because not all instruments could address the science theme of each DG equally, it was decided, in each DG, to introduce the concept of Prime and Secondary instruments. Prime instruments are the

Table 2

List of prime and secondary instruments for the 4 Rosetta Disciplines.

Discipline Group	Rosetta Prime Goals	Rosetta Payload, prime instruments	Rosetta Payload, secondary instruments
DG1	Global Characterisation of nucleus, determination of dynamic properties, surface morphology and composition	OSIRIS, VIRTIS, CONSERT, RSI, MIRO, ALICE	RPC
DG2	Chemical, mineralogical and isotopic compositions of volatiles and refractories in a cometary nucleus.	ROSINA, VIRTIS, ALICE, COSIMA, MIDAS, PHILAE	OSIRIS, MIRO, GIADA
DG3	Physical interrelation of volatiles and refractories in a cometary nucleus.	COSIMA, ROSINA, OSIRIS, MIDAS, GIADA, PHILAE	MIRO, VIRIS, ALICE
DG4	Study the development of cometary activity and the process in the surface layer of the nucleus and in the inner coma (dust-gas interaction)	OSIRIS, VIRTIS, ROSINA, PHILAE, MIDAS, GIADA, ALICE, MIRO, RPC, RSI	COSIMA

instruments addressing by their measurements the science theme of the DG directly, while secondary instruments are considered in a supporting role. Each DG is alternatively in charge of the design of a mission segment, starting from the decision on the trajectory to be flown to establishing the observation timeline to be executed on this segment.

The instrumental participation to the DGs with their Prime and Secondary status was decided during a meeting with the SWT in September 2012, and is detailed in Table 2. A detailed description of the Rosetta instruments can be found in [8].

3.2.1.5. Project science. The ESA Project Scientist acts as the interlocutor between the science community (primarily represented by the SWT) and ESA, and is the chair of the SWT. The NASA project scientist represents the NASA funded PIs and NASA scientists and works closely with the ESA Project Scientist and the RSGS liaison scientists in ensuring the scientific productivity of the mission.

3.2.1.6. The Rosetta science ground segment - liaison scientists. The Liaison scientists are part of the ESA Rosetta Science Ground Segment, acting as the interface for the science planning between ESA and the external components (SWT, PI teams and Discipline Groups). They are coordinating the Discipline Groups and chairing the Discipline Groups discussions.

3.2.1.7. The Rosetta science ground segment - trajectory design and simulations group. The trajectory and simulation group is also part of the ESA Rosetta Science Ground Segment. It is in charge of implementing the high level geometry requirements provided by the Discipline Groups and provide draft trajectories meeting both science and navigation requirements. This group is also in charge of the final trajectory request (TRAL) to the RMOC.

3.2.2. Science planning process summary

The Science Planning process is described in Fig. 1.

The science planning includes the following steps:

- The SWT, supported by the DG members and led by the Project Scientist and the RSGS, agrees on a segmentation per DG of the trajectory for the considered LTP.
- The activity group suggests a ‘Preferred’ comet activity level. This value is then endorsed by the SWT (step (0a) in Fig. 1).
- The RMOC Flight Dynamics team provides the trajectory building blocks available for trajectory design (step (0b) in Fig. 1).
- Based on this activity level estimate and on the scientific requirements provided by the SWT (step (0c) in Fig. 1), a draft trajectory is designed by the RSGS Trajectory and simulation Group (step (1) in Fig. 1).
- For a two weeks’ segment, the Liaison Scientist collects observation

requests from the DG members. All inputs are collected using a large customized DG spreadsheet (step (2)).

- Harmonization of the plan and conflict resolution is done through DG telecons led by the liaison scientist and involving the DG members (step (2) in Fig. 1).
- If prime observations cannot be addressed with the current geometry configuration, feedback is provided to the RSGS Trajectory and simulation Group and the draft trajectory is modified (steps (3a) and (4) in Fig. 1).
- Closer to execution, the Activity Group provides an updated ‘Preferred’ activity level (step (3b)) and the RSGS Trajectory and simulation Group updates the draft trajectory accordingly (step (4)).

The output of this activity is twofold (step (5) in Fig. 1):

- A Trajectory request at Long term planning level (TRAL) submitted to the RMOC 16 weeks before the start of execution.
- A high-level conflict free observation plan, in terms of pointing, data volume and power, with a 6 hours’ granularity.

The TRAL response (LTP trajectory) is then delivered back by the RMOC 4 weeks later (Execution start –12 weeks).

3.3. The medium term planning

The MTP planning cycle starts at execution minus 12 weeks, once the frozen reference trajectory provided by Flight Dynamics in response to the TRAL is available. The objective of the MTP cycle is the generation of the consolidated attitude request. The output of this activity is a Pointing Timeline Request (PTRM) file, which includes all pointing activities related to science observation to be performed by the spacecraft (within trajectory and spacecraft constraints) over a 4 weeks’ period, and a draft of the instrument operation timelines.

3.4. The short term planning

The STP planning cycle starts at execution minus 3 weeks, once the frozen MTP attitude provided by RMOC in response to the attitude request (PTRM) is available. The objective of the STP cycle is the generation of a validated command timeline covering 1 week of payload operations and that will remain within the available resources (data volume and power) [6].

4. Science planning detailed process

We describe hereafter in details the steps of the science planning, up to the end of the LTP phase.

4.1. Trajectory segmentation

The Rosetta science planning starts with the segmentation of the mission per science discipline.

This activity is led by the Project scientist and involves the Discipline Groups, the SWT and the Liaison scientists.

The mission is divided into 2-weeks segments. Each Discipline is given a high priority for a segment, and the associated discipline group members should provide their trajectory requirements to better address their objectives during this period. The segment duration was decided as a good compromise to limit the time gap between two consecutive segments ‘owned’ by the same DG while ensuring that its duration was long enough to complete a trajectory arc (e.g. Close flybys, see below).

The overall segmentation of the trajectory is discussed within the SWT for each LTP period (~4 months), and the distribution of segments per discipline is driven by the science needs for each discipline. The criteria considered can be summarized as follows:

- Main science objectives to be addressed during specific times of the mission
- Time separation between 2 segments for a given prime discipline
- Navigability constraints evolution along the mission, e.g. zero phase angle flybys (for surface imaging, only doable above 2.1 AU) or minimum approach distance
- Optimization of the transition between successive segments to limit the time spent transitioning from one segment to the next.

4.2. STEP 0a - Comet activity level (gas production rate)

4.2.1. Highest and lowest activity level ranges

The design of the trajectory strongly depends on the environment conditions. Before March 2015 (see Section 5), it was expected that the gas production rate was the key parameter to consider to assess navigability: indeed, the relative velocity required to fly at a given altitude depends on the level of activity: the higher the activity level, the faster Rosetta had to fly to avoid increasing its position uncertainty, which would have led to jeopardization of navigation images. Therefore, a key point for the trajectory design is the estimate of the activity level uncertainty (rather than the activity itself).

In December 2011 (~2.5 years before arrival at the comet), the project scientist, supported by the Rosetta science community, provided ranges for the expected gas activity of the comet at various heliocentric distances. Those inputs were requested by the flight dynamics team at the RMOC to support landing activity preparation.

An estimate of the production rates for H₂O, CO, CO₂ was provided based on results from previous ground observations and modelling.

The expected range of water production rate at perihelion was derived from OH and CN measurements [9].

CO₂/H₂O ratios were available in the literature for 10 Jupiter Family comets and 7 Oort Cloud comets ([9,10]). The former included 67 P (at 1.85 AU=7.1%). For the Jupiter Family comets an average ratio of 8% appears to be constant until 2.7 AU. A nominal value of 3–8% CO₂ was chosen for 67 P. CO was considered to be 1–5% of water at perihelion.

A $1/r^2$ dependence was assumed for both, CO₂ and CO. The radial dependence for H₂O was more complicated as a strong decrease was observed beyond 2.7 AU. However, as the water production rate was rather well known at short heliocentric distances, models were used as guidance.

From this, an estimate of the total gas production rate (H₂O, CO, CO₂) at perihelion was derived, and range values for the minimum and maximum expected gas production rates at various heliocentric distances were established (see Table 3; for comparison, production rates derived from Rosetta measurements in 2014–early 2015 are also included, see right column).

Finally, using dust observations made while the nucleus was at various heliocentric distances ([11–15]), and assuming a near-constant dust/gas ratio for optically visible particles (~1 μ m), a confidence check

Table 3

Minimum and Maximum estimated gas production rates in the year 2012. Right column: Measured gas production rates (year 2014–early 2015) are given for comparison.

		min	max	Measured Production rates (2014–2015)
Rh(AU)		molecules/s	molecules/s	molecules/s
Perihelion	Q(H ₂ O)	4.00E+27	1.00E+28	
	Q(TOTAL)	4.14E+27	1.13E+28	
2	Q(H ₂ O)	4.00E+26	2.00E+27	
	Q(TOTAL)	4.60E+26	2.50E+27	
3	Q(H ₂ O)	1.00E+25	3.00E+26	Q (H ₂ O) ~8E+25 (between 2.9 and 2.5 AU) [16]
	Q(TOTAL)	3.70E+25	5.40E+26	Q (total) ~1E+26 (between 2.9 and 2.6 AU) [17]
3.5	Q(H ₂ O)	1.00E+24	2.00E+26	Q(H ₂ O) =4.00E+25 (3.5 AU) [18]
	Q(TOTAL)	2.10E+25	3.80E+26	

that the derived gas production rate values were realistic could be performed.

As shown in Table 3, the estimated gas production rates range is rather large. Moreover, the evolution slope of the gas production rate w.r.t heliocentric distance for the ‘Highest Activity’ and the ‘Lowest Activity’ cases is not parallel. As a precise gas production rate for science planning could not be estimated unambiguously at the time the Long Term science plan would kick off, it was decided to follow the approach of dual planning.

4.2.2. Dual planning implementation

The dual planning consists of the development of 2 science plans in parallel, the so-called ‘Preferred Plan’ and the ‘High activity plan’. The ‘Preferred Plan’, is, as its name suggests, the plan that is expected to provide the best science return. However, this plan is bearing the risk to not be feasible in terms of navigation, would the ‘bet’ on cometary activity value and its evolution over the LTP period (usually 4 months long) have underestimated the true activity. In turn, the high activity plan is considered as a safe fall back, would the spacecraft have to depart from the preferred plan because of navigation issues.

4.2.2.1. Preferred activity level decision. Nominally, the trajectory and Activity Group (through teleconferences chaired by the Project Scientist) makes predictions regarding the activity level and activity profile (slope) to be expected for the LTP planning period under discussion. This estimate is based on earlier Rosetta and ground-based measurements as well as using modelling output. The activity prediction can be derived in two ways:

- A direct assessment of the position uncertainty driven by activity (by considering the actual position errors), which we represent as a percentage of either the Lowest Activity or the Highest Activity level.
- A prediction of the magnitude of the activity (as a percentage of either the Lowest Activity or the Highest Activity level) and then decide on a factor by which the uncertainty is related to activity.

Given the short-term variations of the near nucleus environment, the Activity Group adopted the approach that the gas drag uncertainty should correspond one to one to the magnitude of the gas drag itself (production rate [kg/s] x velocity [m/s]).

4.2.2.2. High activity case plan. As mentioned, a second plan is developed assuming the highest gas production rate values, as

established in Table 3 above. This is referred to as the ‘robust’ (or ‘back up’) plan, i.e. the plan to be flown if the production rate ends up to be above the one predicted by the Activity Group.

The High Activity Case Plan (HAC) is developed by the Discipline Groups and the RSGS, in parallel with the preferred one. It covers trajectory design down to detailed pointing plan (PTRM). It should be noted that due to the high workload involved for all parties, it was decided not to pursue dual planning down to commanding level.

In practice, the RMOC team always attempts to fly the TRAL corresponding to the preferred case, as long as the observed pointing errors are within the limit of 2.5° during navigation data acquisition periods (see Section 4.3 for details). As soon as the pointing errors exceed this limit, a manoeuvre would be executed to move the spacecraft to the ‘robust’ trajectory. During the time the TRAL scheme was applied, this never happened.

4.3. STEP 1 - Trajectory design

Based on the Discipline groups high level trajectory requirements and an estimate of the comet activity level, the RSGS Trajectory design and simulations group starts by creating an approximated implementation of the trajectories (for both the preferred and the HAC cases). This is needed to check constraints and to study the scientific feasibility of the trajectory. The approximate trajectory is also used to initiate preliminary planning of observations. It is approximate in the sense that it only represents Keplerian motion and does not take into account disturbing forces like gravitational effects of the non-spherical shape of the comet, comet outgassing, and solar radiation pressure. Also, the manoeuvres are only modelled approximately.

The trajectory described in a TRAL consists of a sequence of trajectory arcs. There are three possible types of arcs:

- Terminator: A circular orbit, not tilted more than 30° from the terminator plane (optical navigation constraint).
- A far fly-by (FarFB), described by its start and end points.
- A close fly-by (CloseFB), described by its start point and its closest approach point (CAP).

4.3.1. Trajectory design constraints

For each of these types of arc, there are constraints to ensure spacecraft safety and the navigability of the trajectory. These constraints entail distance from the comet, spacecraft velocity, phase angle, and uncertainty of spacecraft position due to comet outgassing:

i. Distance constraints:

The distance to the comet centre must always be larger than 8 km. This must also hold in the case of any manoeuvre interruption and manoeuvre over-performance of up to 50%. For a farFB, the distance to the comet must always be larger than 50 km, also in the case of manoeuvre interruptions. For a terminator arc (arc at phase angle c.a. 90°), the distance must be smaller than 30 km.

ii. Velocity constraints:

For far and close flyby arcs, the spacecraft velocity relative to the comet must always be larger than 20 cm/s, also in the case of any manoeuvre interruption. The spacecraft velocity must also be larger than two times the escape velocity at the pericentre of the arc.

iii. Phase angle constraints:

The phase angle (i. e. the angle Sun–comet–spacecraft) must always be smaller than 120° in order to assure sufficient visibility of the nucleus for optical navigation. Below a solar distance of 2.2 AU (resp. 1.75 AU), this angle must be larger than 40° (resp. 50°). These limits are due to illumination constraints on the $-Z$ face of the spacecraft during mandatory nucleus pointing. These phase angle

constraints must also hold in the case of any manoeuvre interruption.

iv. Pointing error constraints:

During navigation data acquisition periods, the position error must be small enough so that the consequential pointing error is smaller than 2.5° (corresponding to half the field of view of the navigation cameras, NAVCAMs). The position error is computed by integrating the uncertainty of the comet gas pressure on the spacecraft along the trajectory, where the actuation of the solar panels has to be taken into account. In addition, worst case manoeuvre errors have to be considered.

There are two different types of position errors which manifest themselves at different planning phases.

About one month after the delivery of the TRAL, the RMOC responds by delivering the actual, accurately modelled trajectory at LTP level to the RSGS. Based on this trajectory, the planning by the RSGS proceeds with fixing the spacecraft attitude at MTP and the instrument commanding at STP. At VSTP, the abstract description of spacecraft attitude provided by the RSGS is expanded by the RMOC into a real inertially referenced attitude using the latest orbit determination.

The VSTP trajectory is different from the LTP trajectory because of manoeuvre errors (which are in the order of 1%) and uncertainty in the cometary gas drag. Because of the very same disturbing factors, the actually flown trajectory is again different from the VSTP trajectory.

The attitude uncertainty at MTP is caused by the difference between the planned trajectory at VSTP level and the planned trajectory at LTP level. The pointing error relevant for navigation and observation is caused by the difference between the actually flown trajectory and the planned trajectory at VSTP level.

The latter position error — i. e., the spacecraft is not exactly at the location for which the spacecraft attitude was computed — translates into a pointing error for observations of the comet. This pointing error determines the navigability of a trajectory, because if it is too large, navigation images will not contain the comet any more. An example for the experienced pointing error evolution is included in [19].

4.3.2. Trajectory implementation

Fig. 2 shows the trajectory flown during the time when the TRAL scheme was applied.

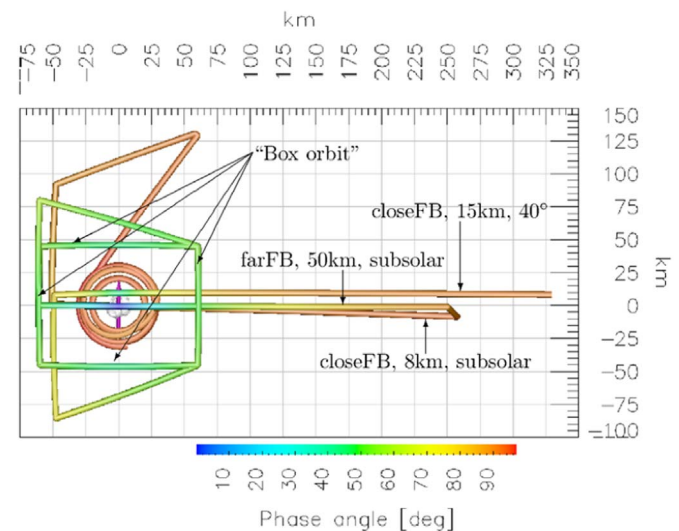


Fig. 2. Trajectory flown by the Rosetta orbiter from 2014-Nov-22 to 2015-Mar-31. The trajectory is shown in the terminator plane, i. e., the view is from the Sun. The rotation axis of the comet (shown as magenta arrow) points upward (projected in the terminator plane). The size of the comet is five times exaggerated.

The trajectory starts with bound orbits of varying radius in the terminator plane. These are favoured because they facilitate staying relatively close to the comet for an extended period of time, to identify gaseous species around the comet and map their abundance as well as to get good grain statistics for grain shape analysis, grain composition, size and velocity distribution. A drawback of bound orbits is that they can only be tilted from the terminator plane by a maximum of 30° because of stray light constraints of the NAVCAMs. Therefore low phase angles are not accessible.

It was assumed that the spacecraft would have to leave bound orbits at some point because of increasing comet activity, and when the uncertainty of comet outgassing began to compromise safe navigation. Then only relatively fast fly-bys would be possible, so that the spacecraft comes close to the comet for a short time only. However, even before this point was expected to be reached, the decreasing solar distance would not allow the spacecraft to point to the comet with a 0° phase angle anymore because of thermal constraints.

Therefore, it was decided to leave the bound orbit on 4th February 2015, just in time to perform two Fly-Bys (further abbreviated to FB) over the subsolar point — one closeFB and one farFB — before the window for this opportunity closed, for e.g. phase function measurement to study bidirectional reflectance of surface materials. These two subsolar fly-bys were designed to be as close as possible to the equatorial plane of the comet — i. e., they are horizontal in Fig. 2—in order to maximize the possibilities for lander communication and CONSERT operation ([20]). The exact landing site was not known when the orbiter trajectory was designed, but it was assumed that it would be close to the equator [2]. Thus, with the lander antenna pointing straight upward, the communication chances would be best with the orbiter flying over low latitudes.

A closeFB inevitably takes the spacecraft far away — several hundred kilometers — from the comet afterwards. The closeFB has to be relatively fast (1–3 m/s) for two reasons: First, a slow closeFB would spend more time close to the comet, which increases the position error due to gas drag uncertainty. Second, the data cut-off following the closeFB would occur at lower distance, where the angular position error (which manifests itself as a pointing error) is larger. After closest approach (CAP), enough time to reliably acquire navigation images has to be allowed for before any subsequent manoeuvre.

After the subsolar fly-bys, the spacecraft goes into a “box orbit”, which is made of rectangular farFBs, cf. Fig. 2. The sides of the box are of different lengths, because manoeuvres always occur at the beginning of a VSTP, and the VSTPs alternate in length between three days and four days. The orientation of the “box-orbit” was chosen to be rather “landscape” than “portrait” in Fig. 2 in order to minimise excursions to high latitudes.

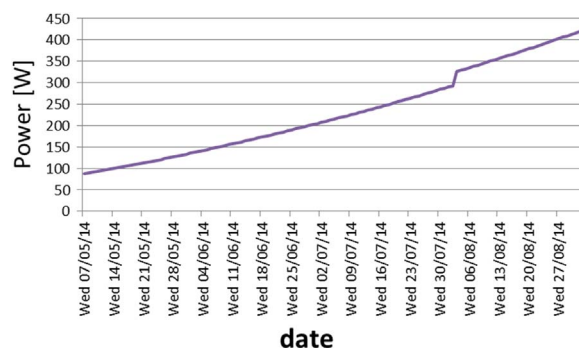


Fig. 3. Power available for payload operations (in Watts) at the start of comet phase until September 2014.

After the “box-orbit”, another closeFB is performed. The minimum possible phase angle due to thermal constraints is 40° . The closeFB is designed to be co-rotational at the CAP. Therefore, it is also in an East-West direction, as favoured for lander communication. Fly-bys planned for later in mission were mostly designed to be in a North-South direction, as this provides better coverage and could also allow a short term phasing in order to fly over a specific point on the comet. However, the TRAL scheme was abandoned before any such fly-by was flown, cf. Section 5.

4.4. Observation constraints

The mission constraints are not limited to the trajectory and pointing, but also to downlink capacities (strongly dependent on geocentric distances) and limited power available (at large heliocentric distances).

4.4.1. Power

When Rosetta woke up in January 2014, it was still at about 4.5 AU from the Sun.

By the time the spacecraft and payload re-commissioning were completed, science observations could be resumed. The large heliocentric distance (4.07 AU in early May 2014) drastically impacted the science operations, and a sequential observation scheme had to be put in place. The available power remained a limiting factor for the first 3 months of science observations.

Fig. 3 shows the evolution of the power available in Watts for payload observations (best case). The jump in the trend is due to the modified spacecraft configuration (from using 2 to 1 Inertial Measurement Package only). For information, the maximum power used by the payload during cruise phase operations was about ~380 W.

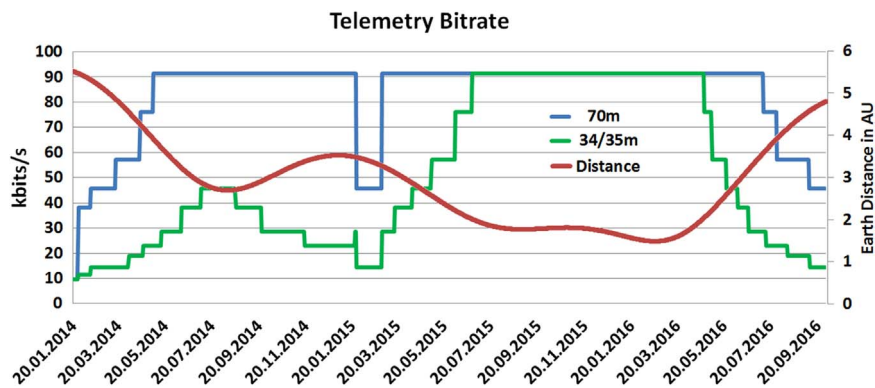


Fig. 4. Red: Rosetta–Earth distance for the years 2014–2016. Blue/Green: Bitrate information available at the time of LTP level science planning. Blue: 70 m DSN-antennas. Green: 34/35 m antennas. The low bitrates in February 2015 are due to a solar conjunction. [In practice the bitrates used with 34/35 m antennas were adapted to the performance of individual stations starting from the end of 2014. However, due to the lead time of LTP level science planning, the revised/improved values are only considered here for the period with decreasing bitrates in 2016. For simplicity the medium performance values of the Cebreros station are used.](For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

4.4.2. Data volume resources and distribution

The geocentric distance of the Rosetta spacecraft changes significantly over the period from hibernation exit until the end of the mission. As a consequence, the telemetry bitrate and hence the data volume available for the scientific instruments varies substantially over the pre-landing and comet escort mission phases (see Fig. 4).

At the time of LTP level planning the detailed pass allocations - in particular for ground stations of the Deep Space Network (DSN) - are normally not yet known. For science planning purposes the available data volume is therefore determined from the bitrate and using an estimate of the effective downlink time per day. For most of the mission periods the total downlink time is assumed to be 16 h per day, which corresponds to an almost complete pass coverage with three different ground stations per day. The resources required for spacecraft and instrument housekeeping telemetry as well as for acquiring navigation images are subtracted in order to derive an estimate of the amount available for instrument science data. At maximum bitrates this results in almost 600 MBytes per day. However, during periods with low bitrates, such as at the end of 2014 or during the solar conjunction in February 2015, the available data volume was scarce and significant resource conflicts between instrument teams occurred. The bottlenecks were partially mitigated by requesting an increase of 70 m antenna support from the DSN.

For data volume allocation purposes the instruments were divided in three groups according to their level of resource usage:

- Group 1: For a number of instruments the data volume is small and limited by technical or operational constraints. These instruments are allowed to acquire all the data they request. The group comprises the following instruments: CONSERT, COSIMA, GIADA, MIDAS, PHILAE, SREM.
- Group 2: A second group of instruments comprising ALICE, MIRO, ROSINA, and RPC typically generates a substantial amount of data and operates (almost) continuously. Guaranteed data rate envelopes are allocated to these instruments, but the values need to be adapted to the available resources and restricted during some mission phases. For periods of maximum available telemetry bitrate (from mid-2015 until ~ Mar 2016, see Fig. 4) these instruments can be operated at their highest data rate.
- Group 3: This group includes the OSIRIS and VIRTIS instruments which produce a very large amount of data in a non-continuous manner. These instruments obtain the largest fraction of the data volume, but do have to accept significant restrictions in some mission periods.

In general, the following rule is applied for distributing data volume during planning periods with significant resource restrictions: After subtracting the data volume required by Group 1 instruments, 30% of the remaining resources are allocated to Group 2 and 70% to Group 3 instruments. Typically, the OSIRIS and VIRTIS instruments then divide their share equally.

4.5. STEP 2- Observation plan development

Once a draft trajectory is available, the detailed development of science observation sequences can start. Both the Preferred Plan (corresponding to the trajectory adapted to the ‘preferred’ activity level) and the High Activity Plan are developed in parallel as the final trajectory request to the RMOC needs to happen at the same time for both cases.

For each 2-week segment, the RSGS provided information regarding the trajectory geometry and expected pointing uncertainties with a 6hrs granularity, and the Discipline Group members are requested to provide their observation requests through the use of a customized spreadsheet.

A distinction is made between observations addressing the prime

discipline science objectives for this segment and the ones belonging to other disciplines. Also, multi-instruments observation campaigns addressing the same science objective are designed so as to maximize the science return.

4.5.1. Customized segment spreadsheets

Following the example of the Titan working groups, implementing the Cassini mission Titan’s flybys (T. Ray, personal communication), observation requests from the different PI teams are collected and harmonized using a customized spreadsheet, providing a high level overview of the operational constraints and science requests, aimed at flagging, and resolving major conflicts and resource bottlenecks. The Rosetta spreadsheet contains the generic geometrical details for a given segment with a 6 h’ granularity (distance to the comet, phase angle, sub-spacecraft latitude and longitude). Some additional details are provided for specific instruments planning purpose: nadir to limb angular distance (for remote sensing pointing development), Nadir to Sun angle (for plasma consortium pointing development), Solar zenith angle values (for the dust analyser instrument). The estimated pointing error (due to the difference between the actually flown trajectory and the planned trajectory at VSTP level, see Section 4.3.1 for details) is also provided for each block as it is a driver for the scheduling of specific observations with high pointing accuracy needs. Finally, an estimate of the available power and data volume is also provided.

The spreadsheet includes a set of rows per instrument where observation names and requested resources (pointing description, data volume and power) need to be provided by the teams for each of their observation requests. In addition, the spreadsheet includes a reference to the Master Science Plan relevant science objectives addressed by the observations, to keep track of the mission’s achievements.

The use of a spreadsheet allows the high flexibility necessary for combining the parallel observations of different instruments, which requires continuous compromises and parameter tunings between instruments to make it possible for other instruments to ride-along (such as scan velocity, exposure duration...).

4.5.2. Observation pointing design and validation

To help the teams developing their observations strategy, they can use a 3D Visualisation tool (3D_Tool). It is a web front-end interface developed by the RSGS, which provides a quick overview of the spatial and temporal evolution of the Rosetta trajectories. The main page is dedicated to the Long Term Planning (LTP) level and study trajectories visualizations (draft TRAL, TRAL and LTP trajectory provided by the Flight Dynamics team in response to the TRAL). In the Medium Term Plan (MTP) level, it provides information about all MTPs already executed and upcoming ones, including output from the LTP and MTP processes such as pointing details visualisation (PTRM). At the Very Short Term Plan (VSTP) level, it provides reconstructed trajectories for all past VSTP periods, and the prediction for the next VSTPs.

The “Flight Dynamics Tool” is a web front-end interface developed by Rosetta RSGS for assessing the compatibility of pointing request files (PTR) with spacecraft constraints imposed by trajectory and spacecraft performance using Flight Dynamics (FD) web services.

Additionally, the “Flying Potato” software is a GUI tool developed by the RSGS for pointing visualisation and checks. On user request it allows editing PTR blocks, quick constraint checking using Flight Dynamics web services, and the visualisation of the pointing request results on the 3D-Tool viewer.

Further visualisation and analysis tools for supporting the science planning process are presented in [19].

4.5.3. Plan harmonization

4.5.3.1. Pointing allocation. Once all teams have provided their observation requests, the harmonization starts. The term harmonization refers to the process aiming at consolidating a

timeline of observation within the resource envelopes (power, data volume) and free of conflicts (pointing requests).

The observation plan harmonization is performed via teleconferences involving the Discipline Group members (at least one per instrument) and chaired by one of the Liaison scientists.

The harmonization starts with prime pointing identification: each six hours' block is discussed within the Discipline Group, and a prime observation is identified based on its priority (if it addresses one of the objectives of the Discipline which owns the segment) and its associated pointing. The pointing details are adapted whenever possible to allow other instruments to ride along so as to optimize the pointing time available while favouring the development of synergistic observations.

One additional criterion to consider for the overall pointing plan is the need for some in situ instruments (mainly the ROSINA gas spectrometers and the COSIMA dust analyser) to make sure the cometary gas (/dust) flux falls within their field of view (5 degrees for COSIMA) for a sufficient amount of time to allow them to integrate enough and derive the main gas and dust properties of the coma. When the spacecraft is close to the comet (20–30 km), this constraint is conflicting with most of the remote sensing pointing requests addressing the same science objectives, since those latter would mainly need to point at the limb (or even further off-nadir). To implement this request, the concept of cost-function was introduced: high-level estimates of the percentage of time spent within 5 degrees are provided (with some additional factoring criteria such as $1/\text{distance}^2$, solar zenith angle, phase angle) for the full segment. Whenever the cost function value is not satisfying the prime in-situ needs, the pointing plan is re-discussed and tweaked to reach an acceptable balance for all primes.

4.5.3.2. Data volume. Once the pointing plan has been agreed, the teams are asked to update their individual plan to reflect the agreements, together with an estimate regarding the generated data volume.

Then, based on science priority and following roughly the data volume allocation scheme (see 4.4.2), the data volume share is re-discussed (proposals are made by the Liaison scientists when needed) and a general agreement is reached between the different teams.

4.5.3.3. Conflict resolution. During the teleconferences, conflicts resulting from the various individual requests are solved based on the identified science priorities. The observations addressing the prime Discipline's objectives are given highest priority for pointing. Also, data volume shares can be adjusted during a specific segment to ensure that the prime science objectives can be addressed.

From time to time, the teams can be asked to provide a more detailed science justification for their observations to support the discussion.

In case of hard conflict that cannot be solved during the teleconference, the Project Scientist needs to take the final decision, based on overall mission level science goals.

4.5.4. STEP 3a and 4 - Optional trajectory update request

During the development of the observations plan, one of the conclusions may be that the draft trajectory does not allow to fully address the science objectives of the prime Discipline. In this case, the RSGS Trajectory and Simulations Group tweaks the trajectory parameters based on the Discipline Groups feedback. This trajectory update can only happen once during the planning process due to the limited time available (see Fig. 1). Once this is done, a TRAL request is provided to the SWT and to the Mission Manager for approval, and then sent to the RMOC 16 weeks before start of execution.

4.6. Operations plan Implementation (MTP assembly)

At the end of the LTP harmonization process an agreed timeline of instrument observations and the distribution of resources is available (step (5) of Fig. 1). For each of the planned observation types, draft pointing implementations called PTR-snippets are collected from the instrument teams and stored in an observation definition library along with corresponding power and data volume resource estimates.

12 weeks before start of execution, LTP product files delivered by Flight Dynamics are also available. It includes the reference trajectory as requested via the TRAL interface and the corresponding pointing timeline skeleton file (PTSL) which contains the schedule of navigation and maintenance blocks during which the pointing is pre-defined and not available for specification by the science instrument teams.

4.6.1. Hand-over to the operations group

The harmonization of the observation schedule, in particular regarding the distribution of prime pointing time, was achieved with a granularity of six hours in the respective planning spreadsheet, whereas the duration of the pointing intervals between spacecraft activities is typically less.

The process of transforming the spreadsheet agreements into a valid draft Pointing Timeline Request (PTR) is accomplished by means of the ASPEN scheduling software. ASPEN is a tool for automation of planning and scheduling for space mission operations developed by the Artificial Intelligence Group at JPL. It was specifically adapted to support the Rosetta mission planning as described in detail in [21].

The output of the ASPEN scheduler is inspected by the liaison scientists with the help of the timeline visualisation functionalities in the ASPEN user interface. If the agreements from the harmonization exercise were correctly translated, the ASPEN output schedule is transferred to draft versions of the operational request files. At this time the respective planning period is handed over to the RSGS operations group for implementation [6], although throughout the MTP implementation phase, the liaison scientist who led the science plan harmonization for the segment monitors the process and assures the scientific integrity of the result.

4.6.2. Implementation of the HAC pointing plan

For the HAC, implementation of the pointing plan at MTP level was carried out by members of the liaison scientist team with support of engineers from the RSGS. The draft PTR from the ASPEN system was checked for validity and provided to the instrument teams for the insertion of their detailed prime pointing block specifications. In order to reduce the effort for this contingency plan, iterations between teams and RSGS were avoided if at all possible. The final merged plan was checked using the Flight Dynamics constraint checking tool (cf. Section 4.5.2) and then submitted to RMOC.

The HAC operations plan development stopped at that point of the process. In case a transition from the Preferred to the HAC trajectory had taken place, implementation of the HAC segment under construction would have been transferred to the operations group, whereas the scientists then would have focused on optimising the schedules for the upcoming periods and evaluating the possibilities for defining an updated preferred case trajectory for the future.

5. Evolution of the planning scheme

During a close fly-by on 14 February 2015 first problems occurred with the performance of the star tracker software in the dust environment close to the comet nucleus. Operational measures were taken for mitigating the problem during the execution of a second close fly-by on 28 March 2015. However, these measures turned out to be not sufficient. The star tracker software was perturbed by a large number of dust particles appearing as 'false stars' and was not able to determine spacecraft attitude during a prolonged period of time. The situation

was critical and could have led to a loss of contact with the spacecraft.

As a consequence, Rosetta had to move further away from the comet to a safer distance. The HAC trajectory, safe in terms of navigation perturbation due to the comet's outgassing, turned out not to be far enough to guarantee a proper functioning of the star trackers in the denser dust environment as the comet activity increased. The main difference between the gas and dust perturbations is that with a high gas pressure, close trajectories could still be flown at the cost of higher spacecraft relative velocity (to compensate for the gas pressure exerted on the spacecraft), while in the case of the high density and brightness of dust particles, the situation could not be solved by other means than increasing distance from the nucleus until the dust rate reduces again (i.e. after perihelion). The distances foreseen in the planned HAC trajectories for the following months were deemed as not sufficiently safe.

For the following LTP planning period the trajectories had already been defined and the corresponding science observations planning was well advanced (for the Preferred case and the HAC). With the new situation these plans were obsolete. At MTP assembly level the pointing plans had been implemented in detail. Also these plans were obsolete.

From this point in time, and due to the uncertainty regarding the dust environment, the trajectory could no longer be planned months in advance and the science planning scheme had to be adapted: it was modified to allow a more dynamic approach and shorter lead times during planning iterations with the PI teams, the RSGS and the RMOC. The main differences between the two processes (pre and post-star tracker issues) are summarized in Fig. 5.

The new planning scheme required an evolution of the technical infrastructure and processes in place at the RSGS, so that the liaison scientists and operations engineers could support the now shorter lead times during planning iterations.

The evolution of the purely operational aspects are described in details in [6]. In this chapter we focus on the high level science planning part.

5.1. Updated science planning process

The updated science planning steps can be summarized as follows:

- 1– Approximately 13 weeks before start of execution, the SWT, supported by the Project Scientist and the RSGS (and after iteration with the Flight Dynamics experts), provides high level

requirements on the trajectory to the RMOC. These requirements should typically cover one 'LTP period' (from 2 to 4 months) and are based on the science objectives to be addressed during this period. It can for example include:

- Phase angle range (Terminator, day-side or night-side)
- Sub-spacecraft latitude range (equatorial, North or South hemisphere)
- Local time region (morning or afternoon)
- Spacecraft relative velocity

So far the top priority criterion has been expressed as 'As Close As Possible As Soon As Possible' with few exceptions:

- When high priority is given to the low phase arcs (e.g. for mapping the southern hemisphere), a safety factor is applied to the distance (increase by 25% with respect to a terminator arc) to cope with larger expected dust flux.
- Additionally, far excursions (either on the day-side towards the bowshock, or on the night-side towards the plasma tail) are also highest priority at two points in the mission when the comet is still active enough, in order to address the high level science objectives of the plasma consortium.

- 2– ~9 weeks before start of execution on-board, the RMOC provides a reference trajectory fulfilling the high level requirements from the SWT. This reference trajectory is used throughout the science planning process to develop an observation plan. The main difference w.r.t the LTP trajectory provided by the RMOC in the previous scheme is that nucleus distance, sub-spacecraft longitude and absolute pointing uncertainty are no longer representative at this stage, as a result of the unknown dust environment and navigability. As a consequence, the observation plan developed at this stage must be robust enough to cope with late changes in the geometry. The RMOC also provides a pointing skeleton file (PTSL) which defines the pointing blocks for navigation and maintenance and the remaining time available for scientific observations with pointing requests.

From ~9 to 5 weeks before execution start, the RSGS liaison scientists remain responsible for the integration of the Rosetta science observations requested by the PI teams into a high level plan. This planning activity is still handled via a customized spreadsheet which

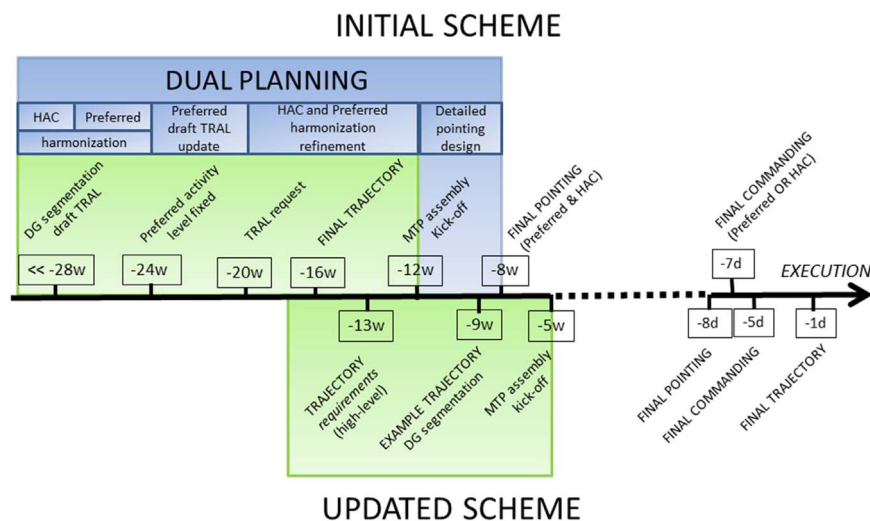


Fig. 5. Comparison between the initial (up) and updated (down) scheme. The main milestones relevant for the science planning are indicated, together with the theoretical lead-in times. The Science planning is highlighted in green, and the Dual planning applied with the initial scheme is indicated in blue. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

allows the liaison scientists and instrument teams to place science observation blocks in time, whilst taking into account geometric constraints as well as resource constraints (ground station availability and predicted data downlink capacity). The spreadsheet is created with a Spreadsheet Generator Tool. Required inputs for the tool are:

- The RMOC pointing skeleton (PTSL) file. Taking into account the PTSL at this stage of the planning prevents from the start to have pointing conflicts that could occur between science observations and spacecraft maintenance or navigation requests, thereby removing one of the tasks the ASPEN scheduler was used for.
- The Pointing Error file to define the estimated pointing error along the MTP period.
- The list of Antenna Outage Exclusion Events which constrain the pointing (in order to guarantee communication windows for critical operations).
- A set of Geometry files (such as those defining the orbit and the attitude of the comet, and the Rosetta reference trajectory provided by the RMOC).

The liaison scientists collaborate with the DG members to harmonize the observations for the different Rosetta instruments, following an iterative process similar to the one in the previous scheme. The main difference is that the negotiation and allocation of observations with pointing requirements is now done based on the available blocks of varying lengths in the pointing skeleton file (PTSL) and not on six hour intervals.

5.2. Updated plan implementation phase

Once the science observations have been harmonized (i.e. 5 weeks before start of execution), the RSGS uses a spreadsheet export tool which allows checking the syntax and consistency of the content and generating operational inputs for the MTP assembly. Science observation schedule events are directly exported into a set of files in a format suitable for the MTP operational software (called MAPPS),

which is then used by the RSGS operations engineers during the implementation and operational validation of the MTP science plan [6].

This direct interface between spreadsheet level science planning and transition to implementation level operational planning replaced the complex ASPEN scheduling tool employed earlier, thereby speeding up the process by leaving out some of the optimization steps previously done by ASPEN.

It has to be noted that with the previous planning scheme, the ASPEN software was used on Rosetta to mainly:

- Support the pointing plan merging (merging of the PTSL and the individual payload pointing blocks)
- Provide an analysis of the data volume at a higher level of granularity than the harmonization spreadsheet.
- Generate the inputs for MAPPS.

With the now reduced lead-in time for planning, the PTSL information can already be accounted for at the start of the harmonization phase, therefore removing the need to merge the different pointing inputs at a later stage. Moreover, the complexity of the ASPEN software (initially developed to be used as a scheduling tool) requires a good understanding of the tool's inputs and outputs, which can be very time consuming. Finally, as the detailed data volume analysis is also done by MAPPS, it was decided to develop more direct and simple interfaces: the spreadsheet generator tool (already including the PTSL information, see above) and the spreadsheet export tool (directly translating the spreadsheet inputs into MAPPS products).

The pointing timeline (PTR) is built using PTR snippets already stored and available for use in the version controlled RSGS observation library. The PTR snippets for the prime observations for all science blocks are merged with the original PTSL file to generate the first version of the integrated PTR. Also originating from the observation definitions, the first version of instrument commanding files is generated and scheduled against observation events as per the original spreadsheet level science plan.

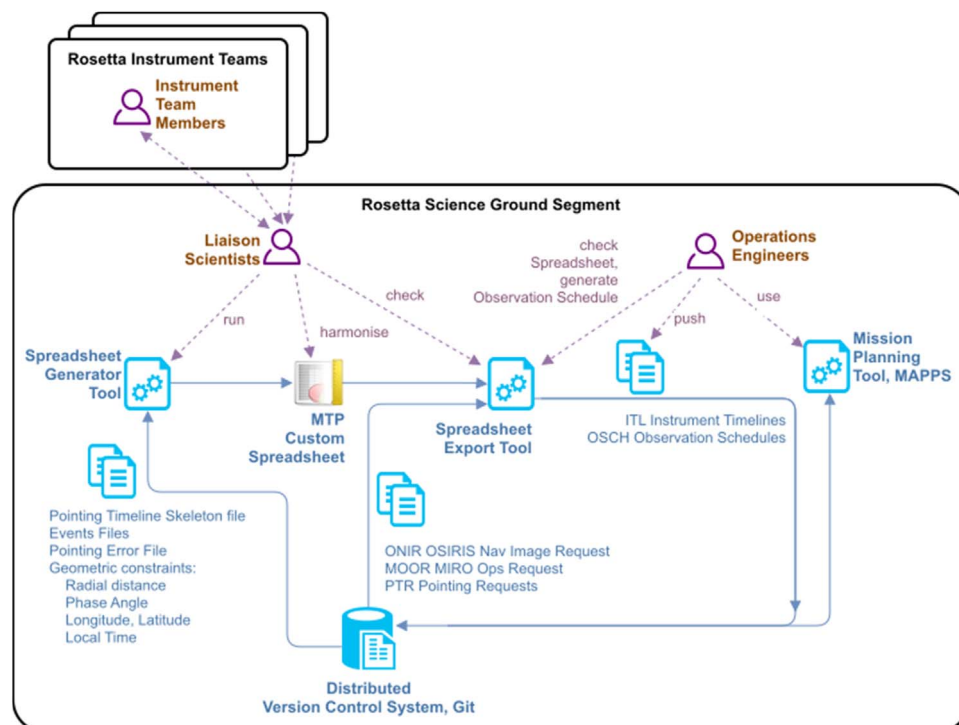


Fig. 6. The new planning implementation process at the RSGS. Developed and deployed in mid-2015 to support the new Rosetta planning scheme after the star tracker issues occurred earlier that year.

These products are then iterated between the RSGS operations engineers and the instrument teams in order to develop and validate the final operational pointing and commanding products that are ultimately delivered to RMOC for final processing and uplink to the spacecraft.

Fig. 6 depicts the updated planning implementation process, starting when a new set of LTP products including the pointing skeleton file is received from the RMOC. The major change of this new scheme, with respect to the TRAL approach, is that the final trajectory is designed by the RMOC on short notice. The short reaction times are required in order to assure safe operations while at the same time approaching the comet as close as possible.

6. Discussion and analysis of results

During the Rosetta comet escort phase, two very distinct science planning approaches have been implemented.

The first one (until March 2015), with its very long lead-in times, proved to be too rigid to cope with an evolving (and partially unknown) environment such as the one of 67 P/Churyumov-Gerasimenko. There were only very limited possibilities to react quickly to unexpected events, and the solution implemented to deal with the environment uncertainties (dual planning) required a huge amount of work by all parties involved (the science teams, the RSGS and the RMOC). Nevertheless, this approach was fully science driven and the feasibility of the science measurements could be tested against the trajectory long before execution. Additionally, the precise knowledge of the trajectory at the time of pointing development allowed to optimize the pointing strategy. In principle, this scheme allows also to implement targeted (landmark) observations, although when approaching perihelion the evolving rotation rate of the nucleus wouldn't have made those pointing feasible anymore. Overall, this planning approach would be suitable for an environment allowing for more predictable trajectories, and future missions to asteroids for example could benefit from such scheme.

The new approach, in turn, is much more flexible from a trajectory point of view, as it can be adapted until the very last minute before uplink. This allows prompt reactivity to the unexpected dust environment, making sure Rosetta can safely approach the nucleus as close as possible. However, as the precise trajectory is not available at the time of Long Term and Medium Term Planning (i.e. at the time of science campaigns and pointing plan development), one needs to ensure that the observation campaigns and pointing remain robust against trajectory changes. Those changes can include altitude, phase angle, sub-spacecraft longitude etc. Additionally, the final pointing plan can only be fully validated at VSTP level, when the final trajectory is available. Therefore, there is not enough time to update the science and pointing plan in case of constraint violations. If such an event happens then the entire VSTP pointing plan is removed (covering 3 or 4 days of pointing) and the pointing skeleton is flown. But this never happened. Finally, targeted observations are not feasible anymore as part of the nominal strategy, unless priorities are shifted from getting 'as close as possible' to targeting specific landmark on the surface for example.

One important point to note is that the Philae Lander did benefit from this updated scheme. In June 2015, after the first Lander communication was received since the end of its First Science Sequence in November 2014, the Rosetta trajectory was rapidly redesigned to increase the chances of communication with the Lander: during several days, Rosetta flew back and forth above the landing site region, maximizing the chance to detect more signal. Unfortunately, this was still not enough to regain a persistent contact with the lander. Early 2016, as the comet flew away from the Sun and the power available to operate the Lander was constantly decreasing, a direct contact became more and more unlikely. Nevertheless, the Lander search continued through imaging with the camera on-board Rosetta (OSIRIS) and was one of many science targets being covered by the instruments onboard. While the Rosetta trajectory remained uncertain until close to execution, a late update of the observation

plan (especially for the pointing) was still possible at that stage, thanks to the new planning scheme in place.

Overall, the original scheme works very well and provides a solid baseline to maximize the scientific output as long as the environment is stable or at least predictable over the planning time scales. On the other hand, for the highly variable comet environment the advantage of a larger flexibility provided by the new scheme outweighs the disadvantage of a much higher level science input to the planning and the risks associated with short turn-around times.

In both cases, having a science plan in place, established through the Discipline Groups process, was extremely beneficial for the overall science return (see [22,23] for example), facilitating high level discussions to agree on campaigns and priorities. It ensures a coordinated implementation and provides a clear route to completing the science objectives of the mission.

7. Conclusions

The science carried out during the Rosetta mission is a huge step toward the understanding of how comets work, and ultimately about understanding how our Solar System was formed.

Having a long term science plan in place has been key to make sure the science objectives of the mission were addressed despite the largely unknown and unpredictable environment.

Nevertheless, the science planning has proven to be highly challenging, thus requiring a constant adaptation to new conditions and a high level of flexibility by the many individuals involved.

In that respect, the Discipline Groups, together with Project Scientists (ESA and NASA) and the RSGS were very successful in coordinating this activity, as demonstrated by the excellent science being done on the observations made possible with the activities described in this paper.

Acknowledgments

Rosetta would not be such an outstanding mission without the work of the many engineers, technicians, and scientists involved in the mission, in the Rosetta spacecraft, and in the instrument teams over the last 20 years whose contributions are gratefully acknowledged. Rosetta is an ESA mission with contributions from its member states and NASA. We acknowledge herewith the work of the whole ESA Rosetta team.

References

- [1] Rosetta Science Management Plan [reference RO-EST-PL-0001]
- [2] M.J. Ashman, et al., Rosetta science operations in support of the Philae mission, *Acta Astronaut.* 125 (2016) 41–64.
- [3] C. Snodgrass, et al., Beginning of activity in 67P/Churyumov-Gerasimenko and predictions for 2014/15, *Astronomy & Astrophysics*, June 4, 2013.
- [4] I. Ferrin, Three predictions: comet 67P/Churyumov-Gerasimenko, comet C/2012 K1 PANSTARS, and comet C/2013 V5 Oukaimeden, *Planet. Space Sci.* 99 (2014).
- [5] J.-L. Bertaux, et al., The water production rate of Rosetta target comet 67P/Churyumov-Gerasimenko near perihelion in 1996, 2002 and 2009 from Lyman α observations with SWAN/SOHO, *Planet. Space Sci.* 91 (2014).
- [6] M. Perez Ayucar et al., Rosetta science operations: The Medium and Short Term Planning, in prep
- [7] B.G. Paczkowski, T.L. Ray, Cassini Science Planning Process, Space OPS 2004 Conference, Montreal, Quebec, Canada, 2004
- [8] K.-H. Glassmeier, et al., The Rosetta Mission: Flying Towards the Origin of the Solar System, Rosetta: ESA's Mission to the Origin of the Solar System, edited by R. Schulz, C. Alexander, H. Boehnhardt, and K.-H. Glassmeier, Springer, New York, Chap. 1, pp. 1–20, 2009.
- [9] L.M. Lara, et al., 67P/Churyumov-Gerasimenko activity evolution during its last perihelion before the Rosetta encounter, *Astron. Astrophys.* 525 (2011) A36.
- [10] T. Ootsubo, et al., AKARI near-infrared spectroscopic survey for CO₂ in 18 comets, *Astrophys. J.* 752 (15) (2012).
- [11] M.F. A'Hearn, et al., Comet Bowell 1980b, *Astron. J.* 89 (1984) 579–591.
- [12] R. Schulz, et al., Rosetta target comet 67P/Churyumov-Gerasimenko, Postperihelion gas and dust production rate, *Astronomy & Astrophysics* 422 (2004) L19.

- [13] C. Tubiana, et al., Comet 67P/Churyumov-Gerasimenko at large heliocentric distance, *Astron. Astrophys.* 490 (2008) 377.
- [14] C. Tubiana, et al., 67P/Churyumov-Gerasimenko at large heliocentric distance, *Astron. Astrophys.* 527 (2011) 113.
- [15] G.P. Tozzi, et al., Evolution of the Dust Coma in Comet 67P/Churyumov-Gerasimenko Before 2009 Perihelion, *Astronomy & Astrophysics*, 2011.
- [16] D. Bockelee-Morvan, et al., First observations of H₂O and CO₂ vapor in comet 67P/Churyumov-Gerasimenko made by VIRTID onboard Rosetta, *Astron. Astrophys.* 583 (2015).
- [17] A. Bieler, et al., Comparison of 3D kinetic and hydrodynamic models to ROSINA-COPS measurements of the neutral coma of 67P/Churyumov-Gerasimenko, *Astron. Astrophys.* 583 (2015).
- [18] S. Gulikis, et al., Subsurface properties and early activity of comet 67P/Churyumov-Gerasimenko, *Science* 347 (2015).
- [19] B. Geiger, Data processing and visualisation in the Rosetta science ground segment, *Acta Astronaut.* 126 (2016) 475–487.
- [20] W. Kofman, et al., The comet nucleus Sounding experiment by Radiowave transmission (CONSERT): a short description of the instrument and of the commissioning stages, *SSR* 128 (2007).
- [21] Chien S. et al., Activity-based scheduling of science campaigns for the Rosetta Orbiter, in: *Proceedings of the 24th International Conference on Artificial Intelligence (IJCAI'15)*, Qiang Yang and Michael Wooldridge (Eds.). AAAI Press 4416–4422, 2015.
- [22] Rosetta mission results pre-perihelion, *Astronomy & Astrophysics* 538, 2015.
- [23] M.G.G.T. Taylor, et al., Rosetta begins its comet tale, *Science* (2015).